Seyed Hamid Hosseini

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Summary

Researcher in reinforcement learning, control, and estimation working on sequential decision-making for systems with hidden state, unknown models, and adaptive adversaries. Designs belief-state, contextual, Bayesian, and game-theoretic methods, implements them end to end, and proves their guarantees, including exploitability bounds with matching lower bounds, convergence to the dynamic-programming optimum, and equilibrium properties. Applications span cyber network defense (CAGE-2), protein engineering (deep mutational scanning), gene regulatory network intervention, autonomous traffic control, and networked multi-agent systems, with industry research experience at Manus Bio.

Education

PhD, Electrical and Computer Engineering, Northeastern University | GPA 4.00/4.0 Jan 2023 – Dec 2026, Boston
Advisor: Prof. Mahdi Imani | Dissertation defended

MSc, Electrical Engineering (Control), Sharif University of Technology | GPA 3.80/4.0 Sep 2018 – Jun 2021, Tehran
Advisor: Prof. Mohammad Saleh Tavazoei | Coursework: Model Predictive, Optimal, Robust, Adaptive, Nonlinear, Multivariable, and Robot Control

BSc, Electrical Engineering, Sharif University of Technology Sep 2014 – Sep 2018, Tehran

Experience

Research Intern | Manus Bio Sep 2026 – Dec 2026, Waltham, MA

- Developed a reinforcement learning approach to protein engineering that frames directed evolution as a sequential decision problem over combinatorial sequence space.
- Implemented the full pipeline in PyTorch: an evolvability-based reward combining experimental measurements with ESM-2 protein language model scores, offline fitted Q-iteration, and a policy-extraction step that uses learned value estimates and attractor-basin analysis to propose experimental candidates.
- Outperformed directed-evolution baselines under matched experimental budgets across several deep mutational scanning landscapes.
- Ran large-scale GPU inference to score millions of variant sequences and established quantitative limits on when language model predictions can substitute for experimental measurement.
- Built a systematic screening framework to identify which experimental designs can support this class of method.

Doctoral Researcher | Imani Research Group, Northeastern University Jan 2023 – Sep 2026, Boston, MA

Contextual RL against strategic adversaries *AAAI 2027, under review*

- Formulated zero-sum partially observable stochastic games with an episode-static latent context, hidden from the defender, that governs the dynamics, observations, rewards, and the adversary's admissible responses.
- Built a nested context–state encoder in PyTorch: a Transformer over the action–observation history identifies the context, a recurrent filter tracks the fast-moving state, and a Soft-InfoNCE contrastive loss with targets from a robust-Bellman pseudometric shapes the context geometry; trained with centralized training and decentralized execution against an adversarial inner loop and a centralized minimax critic.
- Achieved the lowest exploitability among deployable defenders against recurrent PPO, domain-randomized PPO, VariBAD, PEARL, and magnetic mirror descent on four benchmarks (attack graph, dynamic Colonel Blotto, pursuit–evasion, and the CAGE-2 cyber-defense simulator), with 3.8× lower exploitability than the strongest baseline on pursuit–evasion; on CAGE-2, kept ransomware impact active on targeted hosts in only about 11% of rounds.
- Showed through ablations that the design choices carry the gain: the nested encoder recovered 65% of a fully informed oracle's value, versus 32% for a single encoder and 39% when the Bellman-based geometry was replaced by raw-parameter distance.
- Proved that an oracle adversary dominates every private-memory adversary, that a learned representation's one-step reward and prediction errors bound the deployed defense's exploitability with matching lower bounds, that both errors reduce to a single posterior error, and that the robust-Bellman pseudometric bounds the loss of transferring a defense between contexts; on an exactly solvable benchmark, the bound held at every training checkpoint and preserved their ranking (Spearman $\rho = 0.96$).

Belief-based defense against hidden adversary objectives *AAMAS 2027, under review*

- Formulated zero-sum partially observable stochastic games in which the adversary's reward model is drawn per episode from a finite set, known to the adversary and hidden from the defender; showed that the joint posterior over state and reward model is the defender's exact information state, with a contracting minimax Bellman operator.

- Designed a deployable defense that runs the posterior recursion with the co-trained adversary network as its likelihood model, so the filter is exact against the adversary it trains with; trained the defender, adversary, and centralized critic jointly by distilling stage-game equilibria from the critic.
- Least exploitable deployable defender on a Bayesian attack graph (11 nodes, 2,048 states) and a 7×7 pursuit–evasion game, against Shapley iteration with aggregated beliefs, recurrent PPO, domain-randomized PPO, and magnetic mirror descent; 4.4× lower exploitability than the next-best method on pursuit–evasion, where the evader escaped in 50.5% of episodes against this defense versus 76.5% to 96% against the baselines.
- The posterior reached about 0.7 mass on the adversary's true objective within 10 steps from a uniform prior over six candidates.
- Proved an exploitability bound in filter and policy error, each amplified by the effective horizon, with a matching lower bound showing it is order-optimal; the bound ranked likelihood models correctly across training checkpoints (Spearman $\rho = 0.994$).

Bayes-adaptive RL with unknown structure *CDC 2026*

- Learned the dependency structure of binary factored MDPs jointly with the transition dynamics, online, through closed-form Beta–Bernoulli updates, retaining Bayes-optimal exploration without a known factorization.
- Proved a compressed information state sufficient for one-step prediction with bounded suboptimality.

Belief-based RL for asymmetric partially observable zero-sum games *ACC 2026*

- Developed a belief-state actor–critic with centralized training and decentralized execution, and proved local beliefs are sufficient statistics for Nash play.
- Approached the full-information Nash bound on a 10-node network-defense benchmark under up to 50% missed detections.

Deep RL for partially observable regulatory networks *ACC 2025, Best Student Paper Award finalist*

- Designed a belief-conditioned deep Q-network for sequential intervention under partial observability and proved its convergence to the dynamic-programming optimum as observation noise vanishes.

Decentralized RL for asymmetric gene network interventions *TCBB 2025*

- Built a decentralized deep policy-gradient framework for asymmetric intervention against a fully informed adversary; proved robustness guarantees and convergence to the full-state Nash equilibrium under full observability.
- Transferred the method across attacker–defender cyber games and cell–therapy adversarial dynamics.

Game-theoretic modeling of cell–therapy interaction *CAI 2023; CCTA 2024; TCBB 2024*

- Modeled cell–therapy interaction as a two-player zero-sum game with adaptive cellular response, introducing equilibrium interventions for regulatory networks and a partially observed zero-sum Markov game formulation for dynamic intervention.

Bayesian intervention under unknown dynamic stimuli *Information Sciences 2024*

- Designed a Bayesian intervention policy that tracks a posterior over cell responses to unknown dynamic stimuli and selects actions against a set of Nash policies; demonstrated empirical convergence to the optimal Nash policy on p53–MDM2 and melanoma regulatory networks.

Multi-objective intervention synthesis *ACC 2025*

- Formulated multi-objective synthesis with Signal Temporal Logic, trading off worst-case efficacy, response time, intervention frequency, and long-term stability to produce Pareto-optimal policy sets.

Federated RL for autonomous intersections *ICTD 2024*

- Contributed to privacy-preserving federated reinforcement learning for the autonomous control of signalized traffic intersections.

Teaching Assistant | Reinforcement Learning (graduate), Northeastern University Spring 2025, Boston, MA

- Led recitations and graded coursework on MDPs, dynamic programming, policy gradients, actor–critic methods, and deep RL; held office hours and advised student projects.

Graduate Researcher | Tavazoei Lab, Sharif University of Technology Sep 2018 – Jun 2021, Tehran

- Derived the agent-based time-delay margin for event-triggered consensus of networked multi-agent systems, quantifying each agent's tolerance to communication latency.
- Proved the triggering rule is free of Zeno behavior and developed a per-agent algorithm to compute delay tolerance; published in the Asian Journal of Control with N. V. Kuznetsov.

TECHNICAL SKILLS

Programming & Tools: Python (PyTorch, TensorFlow, NumPy, SciPy), MATLAB, Simulink, Git, Linux, CUDA, large-scale GPU inference

Reinforcement Learning: offline RL (fitted Q-iteration), deep Q-networks, actor–critic (CTDE), policy gradient,

multi-agent and decentralized RL, Bayes-adaptive RL, robust and adversarial RL, equilibrium learning

Decision-Making Under Uncertainty: POMDPs and belief-state filtering, approximate information states, partially observed Markov models, stochastic games and Nash equilibria, minimax dynamic programming, Bayesian inference, multi-objective optimization, Signal Temporal Logic

Representation Learning: Transformer encoders, contrastive learning (InfoNCE, Soft-InfoNCE), self-predictive representations, sequential encoders (LSTM, GRU), protein language models (ESM-2)

Control & Estimation: model predictive control, optimal, robust, adaptive, nonlinear, and multivariable control, robot control, event-triggered and networked control, consensus algorithms, recursive Bayesian filtering, system identification

Computational Biology: gene regulatory network modeling, Boolean networks with perturbation, intervention policy design, cell–therapy game modeling, adaptive resistance modeling, benchmark networks (p53–MDM2, melanoma), directed evolution, deep mutational scanning

PUBLICATIONS (80+ citations)

Under review

S. H. Hosseini, S. F. Ghoreishi, M. Imani, and M. Imani, *"Game Approximate Information States: Exploitability Guarantees for Learned Defenses Under Latent Context"*, **AAAI Conference on Artificial Intelligence**, 2027

S. H. Hosseini, S. F. Ghoreishi, M. Imani, and M. Imani, *"Defending Against Hidden Objectives: Belief Filtering with Learned Opponent Models and Order-Optimal Exploitability Bounds"*, **International Conference on Autonomous Agents and Multiagent Systems (AAMAS)**, 2027

Journal articles

S. H. Hosseini and M. Imani, *"Decentralized Reinforcement Learning for Asymmetric Gene Network Interventions"*, **IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB)**, 2025 (link)

S. H. Hosseini and M. Imani, *"Modeling Defensive Response of Cells to Therapies: Equilibrium Interventions for Regulatory Networks"*, **IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB)**, 2024

S. H. Hosseini and M. Imani, *"An Optimal Bayesian Intervention Policy in Response to Unknown Dynamic Cell Stimuli"*, **Information Sciences**, 2024 (link)

S. H. Hosseini, M. S. Tavazoei, and N. V. Kuznetsov, *"Agent-Based Time-Delay Margin in Consensus of Multi-Agent Systems by an Event-Triggered Control Method: Concept and Computation"*, **Asian Journal of Control**, 2023

Conference papers

S. H. Hosseini, F. Zarei, B. Shafai, and M. Imani, *"Bayes-Adaptive Reinforcement Learning in Binary Factored MDPs with Unknown Structure"*, **IEEE Conference on Decision and Control (CDC)**, 2026 (link)

S. H. Hosseini, A. Kazeminajafabadi, A. Kamara, and M. Imani, *"Belief-Based Reinforcement Learning for Asymmetric Partially Observable Zero-Sum Games"*, **American Control Conference (ACC)**, 2026 (link)

S. H. Hosseini and M. Imani, *"Deep Reinforcement Learning for Intervention of Partially Observable Regulatory Networks"*, **American Control Conference (ACC)**, 2025 (link)

S. H. Hosseini, D. Aksaray, and M. Imani, *"Pareto-Optimal Interventions in Gene Regulatory Networks using Signal Temporal Logic"*, **American Control Conference (ACC)**, 2025

S. H. Hosseini and M. Imani, *"Dynamic Intervention in Gene Regulatory Networks: A Partially Observed Zero-Sum Markov Game"*, **IEEE Conference on Control Technology and Applications (CCTA)**, 2024

N. Asadi, **S. H. Hosseini**, M. Imani, D. P. Aldrich, and S. F. Ghoreishi, *"Privacy-Preserved Federated Reinforcement Learning for Autonomy in Signalized Intersections"*, **ASCE International Conference on Transportation and Development (ICTD)**, 2024 (link)

S. H. Hosseini and M. Imani, *"Learning to Fight Against Cell Stimuli: A Game-Theoretic Perspective"*, **IEEE Conference on Artificial Intelligence (CAI)**, 2023

HONORS & AWARDS

Finalist, Best Student Paper Award (1 of 6), American Control Conference (ACC), 2025

Student Organization Leader of the Year (ISAN), Northeastern University, 2026

LANGUAGES

Kurdish (native), Persian (native), English (fluent)